

Student Guide — middle / module-11-multiplication-near-base

IKS Curriculum

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Student Workbook

Student Workbook — Module 11 Nikhilam Multiplication (Middle)

Name: _____ Class: _____ Section: _____

This is your workbook for the next two weeks. Bring it to every class.

Day 1 — Baseline

My baseline time (long multiplication on 97×96): _____ seconds

My first try with Nikhilam: _____ seconds

My target by Day 9: _____ seconds

Day 2 — Deficits from 10

The 9 deficits from 10:

1→9 2→8 3→7 4→6 5→5 6→4 7→3 8→2 9→1

Speed check: can I recall any deficit from 10 in under 1 second? [Y / N]

Practice (Nikhilam, base 10):

Problem	Deficits	Left	Right	Answer
7×8	3, 2	5	6	56
9×8				
6×7				
8×8				

Problem	Deficits	Left	Right	Answer
9×9				

Day 3 — Both Below 100

The procedure (base 100): 1. Find each factor's deficit from 100. 2. Cross-subtract for LEFT. 3. Multiply deficits for RIGHT. 4. Pad RIGHT to 2 digits. 5. Carry overflow into LEFT.

Problem	Deficits	Left	Right (padded)	Answer
97×96	3, 4	93	12	9312
99×98				
95×94				
88×89				

Day 4 — Both Above 100

The change: excesses (positive), cross-ADD instead of subtract.

Problem	Excesses	Left	Right (padded)	Answer
103×104	+3, +4	107	12	10712
108×107				
112×113				

Day 5 — Mixed Cases

The change: one above, one below. Negative RIGHT part.

Problem	Signed	Left	Right (signed)	Answer
102×98	+2, -2	100	-4	9996
104×96				
108×95				

Notice: When excess = deficit, LEFT = base. Then answer = $\text{base}^2 - (\text{deficit} \times \text{deficit})$.

Day 6 — Near 1000

The change: pad RIGHT to 3 digits (base 1000 has 3 zeros).

Problem	Deficits/ excesses	Left	Right (3-digit pad)	Answer
998×997	2, 3	995	006	995006
1003×1004				
970×965				

Day 7 — Why It Works

The algebraic identity:

$$(B - a)(B - b) = B(B - a - b) + ab$$

$$= B \cdot \text{LEFT} + \text{RIGHT}$$

My paragraph (5–7 sentences, base + deficit + algebra):

Day 8 — When to Use It

Use Nikhilam when: _____

Use long multiplication when: _____

Three problems from my daily life that ARE Nikhilam-suitable: 1. _____ 2.

_____ 3. _____

Day 9 — Sprint and Project

Day 1 baseline (long): _____ sec → **Day 9 (Nikhilam):** _____ sec → **Speedup:** _____×

My project format: _____ (video / workbook / data project) **Partner:**

_____ **Featured example:** _____

Day 10 — Reflection

1. What did I set out to make or learn, and what did I end up with?

2. What's the most surprising thing I discovered?

3. Where will I use Nikhilam outside this classroom?

Glossary (your own)

Sanskrit	Meaning	First met
<i>Nikhilam</i>	“all” (short for <i>Nikhilam Navataścaramam Daśataḥ</i>)	Day 1
<i>Nikhilam Navataścaramam Daśataḥ</i>	“all from 9 and the last from 10”	Day 1
<i>Ūrdhva-tiryagbhyām</i>	“vertically and crosswise” (Day 8 mention)	Day 8
<i>sūtra</i>	aphorism / formula	Day 1
deficit	how much a number falls short of the base	Day 1
excess	how much a number exceeds the base	Day 4
base	the power of 10 (10, 100, 1000) used	Day 1

Day 1 — The Hook: 97×96 in Three Seconds

Module: Nikhilam Multiplication · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will record a personal baseline time for 97×96 solved by long multiplication, and will have seen the *Nikhilam* method produce the same answer in under 5 seconds.

Materials

- Printed baseline sheet: a single problem $97 \times 96 = ?$ with workspace and “your time” line. One per student.
- Stopwatch or wall clock with a second hand. One per pair is enough.
- Whiteboard with a single vertical dividing line drawn down the middle (the “Nikhilam split”).

Vocabulary introduced today

- *Nikhilam* (IAST: nikhilam; lit. “all” — short for *Nikhilam Navataścaramaṁ Daśataḥ*, “all from 9 and the last from 10”) — Tirthaji’s 2nd sūtra; here, the rule for multiplication near a base.
- *deficit* — how much a number falls short of the base (97 is 3 short of 100; deficit = 3).
- *base* — a power of 10 the two numbers are close to (here, 100).

Lesson flow

Warm-up (5 min)

- Teacher: “How long does it take you to multiply 97×96 ?” Take 3–4 guesses. Write the range (e.g. “15 sec to 1 min”).
- Teacher: “I’ll show you a way to do it in under 5 seconds. Then we’ll talk about why.”

Baseline timing (10 min)

- Hand out the baseline sheet face-down.
- Students write their name and predicted time on the back.
- On “Go,” flip and solve 97×96 using long multiplication.
- Record actual time. Mark answer; honest self-correction. (Correct answer: 9312.)
- Teacher walks around. Most will take 30–60 seconds. A few faster, a few slower.

Teacher demonstration (10 min)

- On the board, write the problem:

$$\begin{array}{r} 97 \leftarrow (3 \text{ below } 100, \text{ deficit} = 3) \\ \times 96 \leftarrow (4 \text{ below } 100, \text{ deficit} = 4) \\ \hline \end{array}$$

- **Step 1: Find the deficits from 100.** Write -3 next to 97 and -4 next to 96.

$$\begin{array}{r|l} 97 & -3 \\ 96 & -4 \end{array}$$

- **Step 2: Cross-subtract for the LEFT part.** “Take one number minus the *other* number’s deficit. $97 - 4 = 93$. (Equivalently: $96 - 3 = 93$. Either cross gives the same answer.)”
- **Step 3: Multiply the deficits for the RIGHT part.** “ $3 \times 4 = 12$.”

- **Step 4: Stick them together.** “93 on the left, 12 on the right → **9312**.”

$$\begin{array}{r|l}
 97 & -3 \\
 96 & -4 \\
 \hline
 93 & 12
 \end{array}
 \rightarrow 9312$$

- Verify: long-multiplication gives 9312. ✓
- “I never multiplied any 2-digit number by another 2-digit number. I only did $97 - 4$ and 3×4 . That’s the entire trick — when both numbers are near a base.”

Activity (15 min) — First try

- Hand each student a second sheet with four problems:
 - 98×97 , 96×95 , 99×99 , 97×92 .
- Students try the Nikhilam method on each, then verify with long multiplication.
- Time themselves. Expect mistakes and slow times on the first try.

Wrap-up (5 min)

- Show of hands: who got at least one right with Nikhilam? Who was *faster* with Nikhilam already? (Most won’t be — yet.)
- Preview Day 2: “Tomorrow we’ll work on the foundational skill — instantly finding the deficit of any number from 10 or 100.”

Student-facing content

The Nikhilam Method — Multiplication Near a Base

When both numbers are near a power of 10 (10, 100, 1000):

1. **Find the deficit** of each from the base.
2. **Cross-subtract** for the LEFT part (one number minus the *other’s* deficit).
3. **Multiply the deficits** for the RIGHT part.
4. **Concatenate**: LEFT | RIGHT = the answer.

Today’s baseline: my time on 97×96 using long multiplication = ____ seconds. My time using Nikhilam (first try) = ____ seconds.

I expect this gap to FLIP by Day 9.

Homework

- Practice the “complement to 10” reflex: write the deficit of each from 10.
 - $7 \rightarrow$, **8** $\rightarrow$, $9 \rightarrow$, **6** $\rightarrow$, $5 \rightarrow$, **4** $\rightarrow$, $3 \rightarrow$, **2** $\rightarrow$, $1 \rightarrow$ ____.
- Do all 9 in under 10 seconds.

Differentiation

- **Tier up:** Try 103×104 (both *above* 100). Note: instead of cross-subtract, you’ll need to cross-add. See if you can work out the pattern before Day 4.

- **Tier down:** If 2-digit feels too fast, drop to single-digit near 10 (7×8 , 6×9). We'll formally do this on Day 2.

Teacher notes

- The baseline time is the *single most important* artefact of Day 1. Make sure every student records it honestly. Day 9's retest depends on it.
- Don't promise a specific speedup. Say "you'll likely be faster by Day 9; how much depends on you."
- Some students will ask "why does this work?" — defer to Day 7. *"Great question. We'll prove it. For now, just notice that it does work."*
- A few students will try to short-cut and use -3×-4 as the right part, getting a sign-error. Wait until Day 5 for the mixed case. For today, all deficits are positive.

Citation(s) used in this lesson

- *Vedic Mathematics Batch1.pdf* (Vedic Cultural Center, Sammamish WA) — Nikhilam multiplication section.
- *Microsoft Word - sutras1.pdf* — sūtra #2: *Nikhilam Navataścaramaṁ Daśataḥ*.

Day 2 — The Setup: Deficits from 10 (Single-Digit $\times$ Single-Digit)

Module: Nikhilam Multiplication · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will instantly state the deficit of any single digit from 10, AND will solve any single-digit $\times$ single-digit multiplication where both digits are 6–9 using the Nikhilam method.

Materials

- Class wall-poster: "Deficits from 10" — the table $1 \rightarrow 9$, $2 \rightarrow 8$, ..., $9 \rightarrow 1$. Display before class.
- Drill sheet: 20 single-digit $\times$ single-digit problems (both factors 6–9).
- Stopwatch per pair.

Lesson flow

Warm-up (5 min)

- **Complement-to-10 lightning round.** Teacher calls 10 single digits in 30 seconds; class chants the deficit. ($7 \rightarrow 3$. $8 \rightarrow 2$. $6 \rightarrow 4$. $9 \rightarrow 1$)
- Repeat with a different sequence. By the end most students should be calling out the deficit faster than they can think it.

Core (15 min) — Nikhilam at the single-digit level

1. **Set up the template on the board** (use the vertical line):

$$\begin{array}{r|l} 7 & -3 \\ \times 8 & -2 \\ \hline ? & ? \end{array} \quad \begin{array}{l} \leftarrow 7 \text{ is } 3 \text{ below } 10 \\ \leftarrow 8 \text{ is } 2 \text{ below } 10 \end{array}$$

2. **The two operations:**

- LEFT part: cross-subtract. $7 - 2 = 5$. (Or $8 - 3 = 5$. Either cross gives the same number — that's a built-in check.)
- RIGHT part: multiply the deficits. $3 \times 2 = 6$.
- Concatenate: **56**.

3. **Verify.** $7 \times 8 = 56$. ✓

4. **Why does either cross give the same answer?** Quick proof on board: $7 - 2 = 7 - (10 - 8) = 7 + 8 - 10$ $8 - 3 = 8 - (10 - 7) = 7 + 8 - 10$ Same thing. Don't dwell — Day 7 has the full algebra.

5. **The hand-trick (optional, fun).** Hold up both hands. For 7×8 : fold down 2 fingers on the left hand (deficit of 7 from 10 = 3, so keep 7 fingers up; or fold down $7 - 5 = 2$ on the left) and 3 on the right (deficit of $8 = 2$, so fold down $2 + 3 = \dots$ use whichever finger-mnemonic the class likes). The number of fingers DOWN $\times 10 =$ the tens; the number of fingers UP on each hand multiplied = the units. (This is a folk technique that maps to the Nikhilam algebra.) Show once, don't drill it.

6. **Worked examples in chorus:**

- 9×8 : deficits 1, 2. Left: $9 - 2 = 7$. Right: $1 \times 2 = 2$. Answer: **72**. ✓
- 6×9 : deficits 4, 1. Left: $6 - 1 = 5$. Right: $4 \times 1 = 4$. Answer: **54**. ✓
- 6×7 : deficits 4, 3. Left: $6 - 3 = 3$. Right: $4 \times 3 = 12$. Hmm — right part is two digits! Carry the 1 to the left: left = $3 + 1 = 4$; right = 2. Answer: **42**. ✓

7. **Note the carry rule.** When the right part has more digits than the base has zeros (here: base 10, so right part should be 1 digit; if it's 2 digits, carry the extra). Write on board:

"If the right part overflows the base, carry the overflow into the left part."

Activity (20 min) — Drill

Pair drills.

- Hand out the drill sheet — 20 problems, all single-digit $\times$ single-digit, both factors in 6–9.
- Working in pairs:
 - Partner A reads problem 1 aloud.
 - Partner B answers in under 3 seconds using Nikhilam.
 - Swap.

- 10 min of paired drill. Then 5 min of individual silent practice.

After 5 minutes of paired drills, do a class-level **timing test**: project 10 problems on the board; students answer all 10 individually in 60 seconds.

Wrap-up (5 min)

- Quick check: chorus answers — $7 \times 7?$ (49.) $6 \times 8?$ (48.) $9 \times 9?$ (81.) $8 \times 6?$ (48.)
- Preview Day 3: “Tomorrow — both numbers near 100. Same trick, slightly bigger.”

Student-facing content

Single-digit Nikhilam (base = 10)

1. Find each digit’s **deficit from 10**.
2. Cross-subtract for the LEFT part.
3. Multiply the deficits for the RIGHT part.
4. Stick them together. If the RIGHT part is 2 digits, **carry** the overflow.

Examples: $- 7 \times 8$: $7-2=5$; $3 \times 2=6 \rightarrow 56$. $- 9 \times 8$: $9-2=7$; $1 \times 2=2 \rightarrow 72$. $- 6 \times 7$: $6-3=3$; $4 \times 3=12 \rightarrow$ carry 1 $\rightarrow 3+1=4$; right=2 $\rightarrow 42$.

The Sanskrit sūtra: **Nikhilam Navataścaramaṁ Daśataḥ** — “all from 9 and the last from 10.”

Homework

- Practice deficit-from-10 recall. Have a parent/sibling call out single digits 1–9; you answer the deficit. Beat 20 problems in 30 seconds.
- Solve 10 problems on the back of today’s worksheet (all single-digit $\times$ single-digit, both 6–9) using Nikhilam. Self-check against your times table.

Differentiation

- **Tier up:** Try $9 \times 9 \times 9$ (chained) or apply the method to base 20 — see if 19×18 works the same way with base 20. (Yes — deficits 1, 2 $\rightarrow$ left 17, right 2 $\rightarrow 17 \times 20 + 2 = 342$. ✓)
- **Tier down:** Stick to 8×9 , 9×9 , 7×8 for today. Memorise these specific three; we’ll widen the range tomorrow.

Teacher notes

- Some students will protest: “But I already know $7 \times 8 = 56$ from my times tables!” Honest answer: “Yes — for single digits the times-table is faster. The Nikhilam method is here so we can use it on 2-digit numbers tomorrow, where the times-table doesn’t help.”
- Watch for the carry rule on 6×7 and 6×8 . This is the most common error point.
- The “either cross gives the same” property is a built-in check students can use to catch errors. Reinforce it.

Citation(s) used in this lesson

- *Vedic Mathematics Batch1.pdf* (Vedic Cultural Center, Sammamish WA) — Nikhilam multiplication, single-digit examples.
- *Microsoft Word - sutras1.pdf* — sūtra #2: *Nikhilam Navataścaramaṁ Daśataḥ*.

Day 3 — Both Numbers Below 100

Module: Nikhilam Multiplication · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will solve any 2-digit $\times$ 2-digit multiplication where both factors are between 90 and 99 using the Nikhilam method, in under 10 seconds.

Materials

- Whiteboard with the vertical-line template.
- Class wall-poster: “Deficits from 100” — small reference card (e.g. $97 \rightarrow 3$, $88 \rightarrow 12$).
- Pre-printed drill sheet (10 problems).
- Stopwatch per pair.

Lesson flow

Warm-up (5 min)

- **Complement-to-100 lightning round.** Teacher calls 2-digit numbers in the 80s and 90s; class chants the deficit. ($97 \rightarrow 3$. $88 \rightarrow 12$. $96 \rightarrow 4$. $92 \rightarrow 8$. $99 \rightarrow 1$. $85 \rightarrow 15$)
- Repeat. By the end most students should be doing this in under a second.

Core (15 min) — The 2-digit $\times$ 2-digit case, both below 100

1. **The template** (vertical line, base = 100):

$$\begin{array}{r|l} 97 & -3 \\ \times 96 & -4 \\ \hline ? & ? \end{array} \quad \leftarrow \text{deficit from 100}$$

2. **Walk through 97×96** (from Day 1, now formalised):
 - LEFT: $97 - 4 = 93$.
 - RIGHT: $3 \times 4 = 12$.
 - **Pad to 2 digits** (base 100 has 2 zeros): RIGHT = 12 (already 2 digits, no pad needed).
 - Answer: **9312**.
3. **Walk through 98×99 :**
 - Deficits: 2, 1.

- LEFT: $98 - 1 = 97$.
- RIGHT: $2 \times 1 = 2 \rightarrow$ pad to 2 digits $\rightarrow 02$.
- Answer: **9702**.

4. **The padding rule (write on board, boxed):**

Pad the right part to as many digits as there are zeros in the base. Base 10 $\rightarrow$ 1 digit on the right. Base 100 $\rightarrow$ 2 digits on the right. Base 1000 $\rightarrow$ 3 digits on the right.

5. **The overflow / carry case: 88×89 .**

- Deficits: 12, 11.
- LEFT: $88 - 11 = 77$. (Or $89 - 12 = 77$. ✓)
- RIGHT: $12 \times 11 = 132$.
- But the right part should be only 2 digits (base 100). 132 has 3 digits — the leading 1 overflows.
- **Carry the overflow into the left:** left = $77 + 1 = 78$; right = 32.
- Answer: **7832**. ✓ (Calculator check: $88 \times 89 = 7832$.)

6. **The “use either cross” check.** For 97×96 :

- Either $97 - 4 = 93$ or $96 - 3 = 93$.
- If you get *different* numbers, you’ve made an error. This is your built-in error-detector.

Activity (20 min) — Drill

Pair drill, then individual silent drill.

Drill sheet (10 problems, mixed difficulty):

- | | | | |
|-------------------|--------------------|-------------------|-------------------|
| 1. 97×96 | 2. 98×99 | 3. 95×94 | 4. 93×91 |
| 5. 99×98 | 6. 92×95 | 7. 88×89 | 8. 87×92 |
| 9. 96×96 | 10. 91×87 | | |

- **5 min individual:** solve all 10 silently. No talking.
- **5 min paired check:** swap sheets. Partner reads problem 1; the other resolves it on a fresh strip; compare. Continue.
- **10 min mixed work:** problems with both-below-100, individual silent.

Answer key (teacher): | Q | Deficits | Left | Right | Answer | |—|—|—|—|—|—|—|—|—|—| | 1 | 3, 4 | 93 | 12 | 9312 | | 2 | 2, 1 | 97 | 02 | 9702 | | 3 | 5, 6 | 89 | 30 | 8930 | | 4 | 7, 9 | 84 | 63 | 8463 | | 5 | 1, 2 | 97 | 02 | 9702 | | 6 | 8, 5 | 87 | 40 | 8740 | | 7 | 12, 11 | $77+1=78$ | 32 | 7832 | | 8 | 13, 8 | $87-13=79$ or $92-8=84$... | (see note) | 8004 | | 9 | 4, 4 | 92 | 16 | 9216 | | 10 | 9, 13 | $91-13=78$ or $87-9=78$ | 117 $\rightarrow$ carry 1 $\rightarrow$ 79, right 17 | 7917 |

(Note Q8: $87 - 8 = 79$, OR $92 - 13 = 79$. Right = $13 \times 8 = 104$. Carry 1 $\rightarrow$ 80; right = 04. Answer: 8004.)

Wrap-up (5 min)

- **Formative quiz on a slip of paper (5 questions, 3 min)** — see quizzes/formative.md Part A.
- Collect for grading. Hand back tomorrow.

Student-facing content

Both numbers below 100

Step	What you do
1	Find each number's deficit from 100 .
2	Cross-subtract for the LEFT part.
3	Multiply the two deficits for the RIGHT part.
4	Pad the RIGHT part to 2 digits .
5	If the RIGHT part overflows (≥ 100), carry the overflow into the LEFT.

Examples: $-97 \times 96 \rightarrow 93 \mid 12 \rightarrow \mathbf{9312}$ $-98 \times 99 \rightarrow 97 \mid 02 \rightarrow \mathbf{9702}$ $-88 \times 89 \rightarrow 77+1 \mid 32 \rightarrow \mathbf{7832}$ (right part overflowed)

Sanskrit name: *Nikhilam Navataścaramaṁ Daśataḥ* — “all from 9 and the last from 10.”

Homework

- 10 problems on the back of today's worksheet, all 2-digit $\times$ 2-digit with both factors in 85–99.
- Time yourself. Beat the class average from today.

Differentiation

- **Tier up:** Try 83×89 and 78×85 (further from 100). The method still works, but the deficits are larger and the right-part overflow is more common. See how far below 100 you can go before the method stops feeling fast.
- **Tier down:** Stick to factors in 90–99 for the rest of the week. We'll revisit 80s on Day 5.

Teacher notes

- The overflow / carry is the single most common error point. Spend time on 88×89 and 78×85 . If students don't fluent on the carry by end of class, schedule 5 min of catch-up at the start of Day 4.
- The “use either cross” check is a real check — encourage students to do both crosses initially. By Day 6 they'll naturally drop the second one.
- Some students will be fast on a calculator and slow on Nikhilam. Acknowledge: “*With a calculator, calculator wins. The Nikhilam method matters when you DON'T have a calculator — exams, mental estimation, daily life.*”

Citation(s) used in this lesson

- *Vedic Mathematics Batch1.pdf* (Vedic Cultural Center, Sannamish WA) — Nikhilam multiplication, base-100 examples.

Day 4 — Both Numbers Above 100

Module: Nikhilam Multiplication · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will solve any 2-digit $\times$ 2-digit (or 3-digit) multiplication where both factors are slightly *above* 100 (101–112) using the Nikhilam method.

Materials

- Whiteboard with the vertical-line template.
- Day 3 formative quiz returned with marks.
- Drill sheet (10 problems with both factors above 100).
- Stopwatch per pair.

Lesson flow

Warm-up (5 min)

- **Complement-to-100 round** (same as Day 3).
- Add the new round: **excess from 100**. Teacher says “103,” class says “+3.” “108” $\rightarrow$ “+8.” “112” $\rightarrow$ “+12.” “101” $\rightarrow$ “+1.”
- Hand back yesterday’s formative.

Core (15 min) — Both above 100: use excess, cross-ADD

1. **The key change:** instead of *deficit* (negative), we now have *excess* (positive). Instead of cross-subtract, we **cross-add**.
2. **The template** (vertical line, base = 100):

$$\begin{array}{r|l} 103 & +3 \\ \times 104 & +4 \\ \hline ? & ? \end{array} \quad \leftarrow \text{excess from 100}$$

3. **Walk through 103×104 :**
 - Excesses: +3, +4.
 - LEFT: $103 + 4 = 107$. (Or $104 + 3 = 107$. ✓ Either cross.)
 - RIGHT: $3 \times 4 = 12$. Pad to 2 digits: 12.
 - Answer: **10712**. ✓ (Calculator check: $103 \times 104 = 10712$.)
4. **Walk through 108×107 :**

- Excesses: +8, +7.
- LEFT: $108 + 7 = 115$.
- RIGHT: $8 \times 7 = 56$. Pad to 2 digits: 56.
- Answer: **11556**. ✓

5. The pattern:

Both below	Both above
Cross-subtract	Cross-add
Deficit $\times$ deficit	Excess $\times$ excess
Right part is product of POSITIVE numbers (deficits)	Right part is product of POSITIVE numbers (excesses)
<i>In both cases the right part is a positive product. The sign issue only appears in the mixed case (Day 5).</i>	

6. The overflow case (rare here): 112×113 .

- Excesses: +12, +13.
- LEFT: $112 + 13 = 125$.
- RIGHT: $12 \times 13 = 156$. Right part should be 2 digits; the leading 1 overflows.
- **Carry:** left = $125 + 1 = 126$; right = 56.
- Answer: **12656**. ✓

Activity (20 min) — Drill

Drill sheet (10 problems):

- | | | |
|----------------------|---------------------|---------------------|
| 1. 103×104 | 2. 102×105 | 3. 108×107 |
| 4. 109×102 | 5. 101×106 | 6. 111×104 |
| 7. 112×113 | 8. 106×109 | 9. 103×103 |
| 10. 105×105 | | |

- 5 min individual silent.
- 5 min paired check.
- 10 min mixed: alternate problems from yesterday (both below) and today (both above) on the same drill sheet — call it the “mixed-base” sheet. Students need to first **decide** whether each is “both below” or “both above” before applying the right cross-operation.

Answer key (teacher): | Q | Sign | Cross | Left | Right | Answer | |—|—|—|—|—|—|—|—|—|
 —| 1 | both above | + | 107 | 12 | 10712 | | 2 | both above | + | 107 | 10 | 10710 | | 3 | both above | + | 115 | 56 | 11556 | | 4 | both above | + | 111 | 18 | 11118 | | 5 | both above | + | 107 | 06 | 10706 | | 6 | both above | + | 115 | 44 | 11544 | | 7 | both above | + | 125+1=126 | 56 | 12656 | | 8 | both above | + | 115 | 54 | 11554 | | 9 | both above | + | 106 | 09 | 10609 | | 10 | both above | + | 110 | 25 | 11025 |

Wrap-up (5 min)

- **Formative quiz Part B (3 min)** — see quizzes/formative.md.

- Preview Day 5: “Tomorrow — what happens when ONE number is above and ONE is below? Sign care needed.”

Student-facing content

Both numbers above 100

Step	What you do
1	Find each number’s excess above 100 (positive number).
2	Cross-ADD for the LEFT part.
3	Multiply the two excesses for the RIGHT part.
4	Pad the RIGHT part to 2 digits.
5	If the RIGHT part overflows, carry the overflow into the LEFT.

Examples: $-103 \times 104 \rightarrow 107 \mid 12 \rightarrow \mathbf{10712}$ - $108 \times 107 \rightarrow 115 \mid 56 \rightarrow \mathbf{11556}$ - $112 \times 113 \rightarrow 126 \mid 56 \rightarrow \mathbf{12656}$ (right part overflowed)

Same sūtra, mirror operation: when deficits become excesses, subtraction becomes addition.

Homework

- 10 mixed problems (both-below AND both-above mixed). Decide first, then solve. Bring tomorrow.

Differentiation

- **Tier up:** Try 1003×1004 (base 1000). The right part should be 3 digits. Confirm $1003 \times 1004 = 1007012$.
- **Tier down:** Stay with 103×104 , 102×103 , 104×105 until comfortable.

Teacher notes

- The “decide which sign first” step on the mixed drill sheet is the most important habit-builder of this day. Reinforce: *before* you start cross-adding or cross-subtracting, **glance at both numbers** and decide whether they’re both below, both above, or split.
- A few students will try to carry into the right part from the left (instead of the other way). Walk through the place-value reasoning: a “100” overflow on the right is *one more in the next column up*, which is the left part.

Citation(s) used in this lesson

- *Vedic Mathematics Batch1.pdf* (Vedic Cultural Center, Sammamish WA) — Nikhilam multiplication, “above the base” examples.
- *Vedic_Maths.pdf* (ISKCON Desire Tree) — practice problems for above-base multiplication.

Day 5 — Mixed Cases: One Above, One Below

Module: Nikhilam Multiplication · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will solve any 2-digit $\times$ 2-digit multiplication where one factor is *above* 100 and one is *below*, handling the negative right-part correctly.

Materials

- Whiteboard.
- Drill sheet (8 mixed-sign problems).
- `activities/activity-01-mixed-signs.md` worksheet.

Lesson flow

Warm-up (5 min)

- Complement-to-100 round.
- Excess-from-100 round.
- Now mix: teacher says “97,” class says “-3.” Teacher says “103,” class says “+3.” This trains the sign attention.

Core (15 min) — One above, one below

1. **Set up 102×98 .** Write on board:

$$\begin{array}{r|l} 102 & +2 \\ \times 98 & -2 \\ \hline ? & ? \end{array} \quad \begin{array}{l} \leftarrow \text{excess} \\ \leftarrow \text{deficit} \end{array}$$

2. **The left part — either cross still works:**

- $102 + (-2) = 102 - 2 = 100$.
- $98 + 2 = 100$. (Same answer ✓.)

3. **The right part — sign matters:**

- $(+2) \times (-2) = -4$.
- The right part is **negative**!

4. **What does it mean to have a “negative” right part?**

- Concatenating “100” and “-4” gives... $100 \mid -4 = 100 \times 100 + (-4) = 10000 - 4 = 9996$.
- Check: $102 \times 98 = 9996$. ✓

5. **The “borrow” maneuver — cleaner mental version.** Same problem:

- The negative right part of -4 says: “subtract 4 from the natural concatenation $100|00 = 10000$.”
 - So $102 \times 98 = 10000 - 4 = \mathbf{9996}$. Clean.
6. **Or — the “borrow from left” form:** if students prefer always-positive right parts:
- Left = 100. Right = -4 . Borrow 1 from left (which is worth 100 in the right):
 - New left = $100 - 1 = 99$. New right = $100 - 4 = 96$ (now positive).
 - Answer: **9996**.
7. **Worked examples in chorus:**
- 104×96 : excess +4, deficit -4 . Left = 100 (either cross). Right = $+4 \times -4 = -16$. Answer: $100 \times 100 - 16 = \mathbf{9984}$. ✓
 - 97×103 : deficit -3 , excess +3. Left = 100. Right = $-3 \times +3 = -9$. Answer: **9991**. ✓
 - 108×95 : excess +8, deficit -5 . Left = $108 - 5 = 103$ (or $95 + 8 = 103$ ✓). Right = $+8 \times -5 = -40$. Answer: $103 \times 100 - 40 = 10300 - 40 = \mathbf{10260}$. ✓
8. **The pattern (write on board):**

Case	Cross	Right part sign
Both below	Subtract	+ (deficit $\times$ deficit)
Both above	Add	+ (excess $\times$ excess)
Mixed	Either form — still cancels	– (excess $\times$ deficit)

Activity (20 min) — [activities/activity-01-mixed-signs.md](#)

Brief: - 8 problems on the activity sheet, all mixed-sign. - 8 min individual silent. - 7 min paired check. - 5 min: students re-do any they got wrong, OR move to challenge problems if all correct.

Wrap-up (5 min)

- Quick chorus check: $102 \times 98?$ (9996.) $104 \times 96?$ (9984.) $108 \times 95?$ (10260.)
- Preview Day 6: “Tomorrow — base 1000. The pad becomes 3 digits.”

Student-facing content

Mixed case: one above, one below

1. Find the **excess** of the bigger number and the **deficit** of the smaller (or just signed: + for above 100, – for below).
2. LEFT: cross-subtract using signs. $102 + (-2) = 100$. Either cross still works.
3. RIGHT: multiply the two signed numbers. The RIGHT part is **negative**.
4. Subtract the absolute value of the right from LEFT $\times 100$.

Examples: $-102 \times 98 \rightarrow 100 | -4 \rightarrow 10000 - 4 = \mathbf{9996}$ - $104 \times 96 \rightarrow 100 | -16 \rightarrow 10000 - 16 = \mathbf{9984}$ - $108 \times 95 \rightarrow 103 | -40 \rightarrow 10300 - 40 = \mathbf{10260}$

Tip: for $(a+x)(a-y)$ with $a = \text{base}$, the left is $a + x - y$ and the right is $-xy$. Whenever excess equals deficit, the left is exactly the base and the answer is $\text{base}^2 - xy$.

Homework

- 10 mixed problems (mix of both-below, both-above, and mixed-sign). Decide the case first.
- Find ONE example in your textbook where mixed-sign Nikhilam would be faster than long multiplication. Bring it.

Differentiation

- **Tier up:** Try 121×79 (further from 100). Excess +21, deficit -21. Cross: $121 - 21 = 100$. Right: $+21 \times -21 = -441$. Answer: $10000 - 441 = \mathbf{9559}$. (Equivalently: $(100+21)(100-21) = 100^2 - 21^2 = 10000 - 441$.) Notice you've just used the *difference of squares* identity. Connect to algebra.
- **Tier down:** Stay with 102×99 , 103×98 (small differences). Build up the sign care before tackling larger gaps.

Teacher notes

- The mixed case is where most students give up and revert to long multiplication. **Don't let them.** The sign care is genuinely tricky for one day; by Day 6 it'll be a non-issue.
- The “tier up” connection to difference-of-squares ($a^2 - b^2$ identity) is a real bridge to algebra. If a student spots it, celebrate.
- Some students will resist the negative right part. Show them the “borrow from left” form (above) as an alternative — both forms give the same answer.

Citation(s) used in this lesson

- *Vedic Mathematics Batch1.pdf* (Vedic Cultural Center, Samskritam WA) — Nikhilam multiplication, mixed-sign examples.

Day 6 — Near 1000: Padding the Right Part to 3 Digits

Module: Nikhilam Multiplication · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will solve 3-digit $\times$ 3-digit multiplications where both factors are within ~ 15 of 1000, correctly padding the right part to 3 digits.

Materials

- Whiteboard.
- Drill sheet (8 problems near 1000).
- Reference card: “deficits from 1000” (a few worked: $998 \rightarrow 2$, $991 \rightarrow 9$, $988 \rightarrow 12$, $1003 \rightarrow +3$, etc.).

Lesson flow

Warm-up (5 min)

- Mixed sign-and-magnitude round: teacher says a number, class says deficit (if below 1000) or excess (if above).
- $1003 \rightarrow +3$. $998 \rightarrow -2$. $991 \rightarrow -9$. $1012 \rightarrow +12$. $988 \rightarrow -12$.

Core (15 min) — Same rule, bigger pad

1. **The rule is unchanged.** Set up the vertical line, find deficits/excesses, cross-subtract or cross-add, multiply the deficits/excesses.
2. **The ONE change: pad the right part to 3 digits.** Base 1000 has 3 zeros. The right part must be exactly 3 digits.
3. **Walk through 998×997 :**
 - Deficits from 1000: 2, 3.
 - LEFT: $998 - 3 = 995$. (Or $997 - 2 = 995$. ✓)
 - RIGHT: $2 \times 3 = 6$. Pad to 3 digits: 006.
 - Answer: **995006**. ✓ (Calculator: $998 \times 997 = 995006$.)
4. **Walk through 1003×1004 :**
 - Excesses: +3, +4.
 - LEFT: $1003 + 4 = 1007$.
 - RIGHT: $3 \times 4 = 12$. Pad to 3 digits: 012.
 - Answer: **1007012**. ✓
5. **The overflow case: 988×985 :**
 - Deficits: 12, 15.
 - LEFT: $988 - 15 = 973$. (Or $985 - 12 = 973$. ✓)
 - RIGHT: $12 \times 15 = 180$. Pad to 3 digits: 180 (already 3 digits, no overflow).
 - Answer: **973180**. ✓
6. **The harder overflow: 985×978 :**
 - Deficits: 15, 22.
 - LEFT: $985 - 22 = 963$. (Or $978 - 15 = 963$. ✓)
 - RIGHT: $15 \times 22 = 330$. Already 3 digits.
 - Answer: **963330**.
 - Verify: $985 \times 978 = 963330$. ✓
7. **The hard overflow: 970×965 :**
 - Deficits: 30, 35.
 - LEFT: $970 - 35 = 935$.
 - RIGHT: $30 \times 35 = 1050$. That's 4 digits — overflow! Carry the leading 1: left = $935 + 1 = 936$; right = 050.

- Answer: **936050**. ✓
- 8. **Mixed-sign case near 1000:** 1002×998 :
 - Excess +2, deficit -2.
 - LEFT: 1000 (either cross).
 - RIGHT: $+2 \times -2 = -4$.
 - Answer: $1000 \times 1000 - 4 = \mathbf{999996}$. ✓

Activity (20 min) — Drill

Drill sheet:

- | | | |
|----------------------|-----------------------|-----------------------|
| 1. 998×997 | 2. 996×995 | 3. 999×998 |
| 4. 988×985 | 5. 1003×1004 | 6. 1005×1002 |
| 7. 1002×998 | 8. 970×965 | |

- 10 min individual silent.
- 10 min paired check + redo errors.

Answer key (teacher): | Q | Sign | Deficits/excesses | Left | Right (padded) | Answer | |—|—|—|—|
 —|—|—| | 1 | below | 2, 3 | 995 | 006 | 995006 | | 2 | below | 4, 5 | 991 | 020 | 991020 | | 3 | below |
 1, 2 | 997 | 002 | 997002 | | 4 | below | 12, 15 | 973 | 180 | 973180 | | 5 | above | 3, 4 | 1007 | 012 |
 1007012 | | 6 | above | 5, 2 | 1007 | 010 | 1007010 | | 7 | mixed | +2, -2 | 1000 | -4 | 999996 | | 8 |
 below | 30, 35 | $935+1=936$ | 050 | 936050 |

Wrap-up (5 min)

- **Formative quiz Part C (3 min)** — see quizzes/formative.md.
- Preview Day 7: “*Tomorrow we prove why this works. One line of algebra.*”

Student-facing content

Near 1000 — 3-digit pad

The rule is unchanged. The **only** change: pad the right part to **3 digits** (because base 1000 has 3 zeros).

Examples: - $998 \times 997 \rightarrow 995 | \mathbf{006} \rightarrow \mathbf{995006}$ - $1003 \times 1004 \rightarrow 1007 | \mathbf{012} \rightarrow \mathbf{1007012}$ - $970 \times 965 \rightarrow 935+1=936 | \mathbf{050} \rightarrow \mathbf{936050}$ (right part overflowed) - $1002 \times 998 \rightarrow 1000 | -4 \rightarrow \mathbf{999996}$ (mixed)

General rule: the right-part pad-width equals the number of zeros in the base.

Homework

- 8 problems on the back of today’s worksheet, mixed near-1000.
- Compute one of them with long multiplication. How much longer did long take?

Differentiation

- **Tier up:** Try 9998×9997 (base 10000). Right part pad = 4 digits. Confirm answer: 99950006.

- **Tier down:** Stay with the easy near-1000 problems (998×997 , 999×998). Build up.

Teacher notes

- The padding rule is the only new thing today. If students have Day 3 fluent, this lesson is mostly drill.
- Forgetting to pad is the most common error. A student who writes $998 \times 997 = 9956$ instead of 995006 has forgotten to pad to 3 digits. The diagnostic: their answer has the wrong number of digits.
- The “general rule” (pad-width = number of zeros in base) is a real generalisation worth boxing on the board. It sets up Day 7’s algebra.

Citation(s) used in this lesson

- *Vedic Mathematics Batch1.pdf* (Vedic Cultural Center, Sammamish WA) — Nikhilam multiplication, base-1000 examples.

Day 7 — Why It Works: The Algebra

Module: Nikhilam Multiplication · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will state the algebraic identity behind the Nikhilam method $(B - a)(B - b) = B(B - a - b) + ab$, derive it on paper, and explain in their own words why this matches the procedure.

Materials

- Whiteboard.
- The 16-sūtra list (*Microsoft Word - sutras1.pdf*) — printed and visible.
- Worksheet: “explain to a friend” prompt + space.

Lesson flow

Warm-up (5 min)

- Quick Nikhilam round on 97×96 (chorus answer: 9312) and 103×104 (10712).
- “You’ve been doing this for a week. Today we prove why it works.”

Core (20 min) — The algebraic proof

1. **Set the variables.** Call the base B ($= 100$, say). Call the two factors $B - a$ and $B - b$, where a and b are the deficits.

For 97×96 : $B = 100$, $a = 3$, $b = 4$. Then $97 = B - a$ and $96 = B - b$.

2. **Multiply directly:**

$$\begin{aligned}
(B - a)(B - b) &= B^2 - Ba - Bb + ab \\
&= B^2 - B(a + b) + ab \\
&= B(B - a - b) + ab \\
&= B \cdot \text{LEFT} + \text{RIGHT}
\end{aligned}$$

where $\text{LEFT} = B - a - b$ and $\text{RIGHT} = ab$.

3. **Match this to the Nikhilam procedure.**

Algebra	Nikhilam step
$B - a - b$	Cross-subtract: take one number minus the <i>other's</i> deficit
ab	Multiply the deficits for the RIGHT part
$B \cdot \text{LEFT} + \text{RIGHT}$	Concatenate (with the right pad)

4. **Check: cross-subtract gives $B - a - b$.**

- $(B - a) - b = B - a - b$. ✓ (97 - 4 = 93.)
- $(B - b) - a = B - a - b$. ✓ (Either cross gives the same.)

5. **Why concatenation works.**

- The “LEFT” digit pattern, placed in the next-higher slot, means we’re multiplying it by B. (E.g. $93|12 = 93 \times 100 + 12 = 9300 + 12 = 9312$.)
- So $93|12$ is exactly $B \cdot \text{LEFT} + \text{RIGHT} = 100 \cdot 93 + 12 = 9312$.
- That’s exactly what the formula said.

6. **The “above the base” case.** Use $(B + x)(B + y)$:

$$\begin{aligned}
(B + x)(B + y) &= B^2 + Bx + By + xy \\
&= B(B + x + y) + xy \\
&= B \cdot \text{LEFT} + \text{RIGHT}
\end{aligned}$$

Now $\text{LEFT} = B + x + y$ — cross-**add**. $\text{RIGHT} = xy$. Sign flips, procedure mirrors.

7. **The mixed case.** Use $(B + x)(B - y)$:

$$\begin{aligned}
(B + x)(B - y) &= B^2 - By + Bx - xy \\
&= B(B + x - y) + (-xy) \\
&= B \cdot \text{LEFT} + \text{RIGHT}
\end{aligned}$$

$\text{LEFT} = B + x - y$ (signed cross). $\text{RIGHT} = -xy$ (negative). The negative right part is the sign issue from Day 5.

8. **Why the pad-width = zeros in base.** Because $B \cdot \text{LEFT} + \text{RIGHT}$ requires RIGHT to occupy the slots from 0 up to $B - 1$. If $B = 100$, RIGHT is at most a 2-digit number; if it overflows to 100 or more, the overflow rolls into LEFT . That’s the “carry overflow” rule from Day 3.

Activity (15 min) — Explain to a friend

- Worksheet prompt: > Imagine your Grade 4 cousin asks: “Why does that Nikhilam trick work? Is it magic?” In 5–7 sentences, explain why it’s NOT magic. Use the words **base**, **deficit**, and **algebra**.
- 10 min individual writing. 5 min pair-sharing: read your paragraph to your partner; they tell you one part that’s clear and one part that’s unclear.

Wrap-up (5 min)

- Read aloud one strong paragraph (with permission).
- The sūtra connection: write on the board:

“*Nikhilam Navataścaramaṁ Daśataḥ*” — “all from 9 and the last from 10” — is the rule for computing the **deficit** quickly (which we drilled on Day 2). The *multiplication* technique uses that deficit via the algebra above.
- Preview Day 8: “Tomorrow — when is this method actually faster? And when should you use long multiplication?”

Student-facing content

The algebra behind Nikhilam

Let B be the base. Let a and b be the deficits.

$$(B - a)(B - b) = B(B - a - b) + ab$$

- $B - a - b$ is the LEFT part (cross-subtract).
- ab is the RIGHT part (product of deficits).
- The concatenation LEFT | RIGHT equals $B \times \text{LEFT} + \text{RIGHT}$ — which is exactly what the algebra produces.

For excess (both above base): $(B + x)(B + y) = B(B + x + y) + xy$. Cross-ADD; right part is positive.

For mixed: $(B + x)(B - y) = B(B + x - y) + (-xy)$. Negative right part.

The sūtra *Nikhilam Navataścaramaṁ Daśataḥ* names the technique for computing the deficit fast. The multiplication shortcut is the algebraic application.

Homework

- Write a 1-paragraph explanation suitable for a Grade 7 student (your own grade level). Include one worked example with the algebra shown.
- Prove $(B + x)(B - x) = B^2 - x^2$ using the above identity. Find an example where this is faster than long multiplication. (Hint: 101×99 .)

Differentiation

- **Tier up:** Generalise. For three factors $(B - a)(B - b)(B - c)$, expand. Does the Nikhilam pattern extend? (Yes — for three factors near a base, see *Vedic Mathematics Batch1.pdf* extension section.)

- **Tier down:** Skip the formal algebra. Verify the identity by plugging in numbers (B=100, a=3, b=4 → both sides give 9312).

Teacher notes

- The honest framing in one sentence: “*The Sanskrit sūtra names the deficit-computation rule. The algebraic identity is what makes the multiplication trick work, and that identity is just standard FOIL / distributivity.*”
- A few students will object: “*If the algebra is just algebra, why call it Vedic?*” Honest answer: “*Good catch. The procedure has a Sanskrit name and was published in 1965 by Tirthaji. The math itself is older than any name — it’s the distributive property. The Vedic-mathematics framing is contested among historians; we’ll engage with that more at Senior band.*”
- Reinforce: this is *not* a debunking. It’s an honesty about what’s old (algebra), what’s modern (the packaging as 16 sūtras), and what’s contested (the Vedic provenance).

Citation(s) used in this lesson

- *Vedic Mathematics Batch1.pdf* (Vedic Cultural Center, Sammamish WA) — Nikhilam multiplication, “why it works” section.
- *Microsoft Word - sutras1.pdf* — list of 16 Tirthaji sūtras.
- Optional teacher reference: *FUNDAMENTAL AND VEDIC MATHEMATICS .pdf*.

Day 8 — When To Use It: Nikhilam vs. Long Multiplication

Module: Nikhilam Multiplication · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will articulate the regime where Nikhilam is faster than long multiplication, AND will recognise problems where long multiplication (or *Ūrdhva-tiryagbhyām*, the general “vertically and crosswise” method) is the better choice.

Materials

- Whiteboard, two columns labelled “Use Nikhilam” and “Use long / Ūrdhva.”
- “Decide-or-don’t” card deck — 20 multiplication problems on cards. (Teacher prepares ahead.)
- Reflection slip.

Lesson flow

Warm-up (5 min)

- Quick Nikhilam: 97×96 (9312), 104×103 (10712), 102×98 (9996).
- “Now — would Nikhilam work for 47×53 ? Let’s see.”

Core (15 min) — When it works, when it doesn't

1. **The honest claim.** Nikhilam multiplication is fast *only* when both factors are close to the same power of 10.
2. **Try a “bad” example: 47×53 :**
 - Base = ? Neither factor is near 10 (too big), near 100 (too far below), or near 50 (a *sub-base*; advanced).
 - If you force base 50: deficits $+(-3)$, $+3$. Left = $47 + 3 = 50$. Right = $(-3)(+3) = -9$. Answer = $50 \times 50 - 9 = 2491$. ✓ Slow because you have to compute $50 \times 50 = 2500 =$ a *multiplication*.
 - **Sub-bases work** (Day 8 tier-up), but they're not faster than long multiplication for casual users.
3. **A genuinely bad case: 47×89 :**
 - Not near any common base.
 - Use long multiplication. **It's faster.**
4. **The decision rule (write on board, boxed):**

Use Nikhilam when:

 - Both factors are within ~ 5 of a power of 10 (10, 100, 1000): trivially fast.
 - Both within ~ 10 – 15 of a power of 10: still faster than long multiplication for most students.

Use long multiplication (or *Ūrdhva*) when:

 - One factor is “random” (not near a base).
 - The factors are near different bases.
 - Speed doesn't matter (writing out long multiplication is more error-resistant).
5. **The contrast with Module 7 ($\times 11$) and Module 9 (subtraction).**
 - Module 7's $\times 11$ trick works *always* when one factor is 11. It's a different niche.
 - Module 9's subtraction-by-complement (Nikhilam again) works when subtracting from a power of 10.
 - Nikhilam multiplication is just one shortcut in a toolkit. Long multiplication remains the workhorse.
6. **The other Tirthaji sūtra mentioned today: *Ūrdhva-tiryagbhyām*** (“vertically and crosswise”) is Tirthaji's sūtra #3 — the **general** multiplication procedure. It works for any pair of numbers. We won't drill it in this module; it's a topic of its own. Just notice: *Ūrdhva* covers what Nikhilam can't.

Activity (20 min) — “Decide-or-don't” game

Rules: - Pairs receive a deck of 20 cards. Each card has one multiplication problem. - For each card, the pair must **first decide** in 5 seconds: Nikhilam or long? - Then they solve it. - If they used Nikhilam and the method actually applies: full marks. - If they used long multiplication and

Nikhilam would have been visibly faster: half marks. - If they used Nikhilam where it wasn't suitable and got stuck: zero marks (but learning is the lesson).

Sample cards:

97×96 → Nikhilam (both near 100)
 47×53 → Long (no common base)
 998×997 → Nikhilam (near 1000)
 73×84 → Long
 103×104 → Nikhilam
 89×91 → Long-ish (near 90, but no nice base)
 102×99 → Nikhilam (mixed)
 347×28 → Long
 988×994 → Nikhilam (near 1000, larger deficits but still works)
 56×89 → Long
 99×99 → Nikhilam (101×99 would also be nice)
 1003×1004 → Nikhilam
 65×75 → Long
 107×108 → Nikhilam
 1234×9876 → Long (too far apart, too random)
 9999×9998 → Nikhilam (near 10000)
 112×113 → Nikhilam (with carry)
 156×178 → Long
 49×51 → Nikhilam (both near 50 if comfortable; else long)
 997×1003 → Nikhilam (mixed, near 1000)

- 20 cards in 20 minutes. Pairs that finish early swap decks with another pair.

Wrap-up (5 min)

- Class discussion: which cards did pairs most often mis-categorise?
- Insight to leave with: **the method is a tool, not a rule.** Long multiplication is *not* “the wrong way.” Each tool fits a certain shape of problem.
- Preview Day 9: speed sprints + project start.

Student-facing content

When to use Nikhilam — and when not to

Use Nikhilam	Use long multiplication
Both factors within ~10 of 10, 100, 1000, 10000	One factor “random” (not near a base)
Mental math, no pencil	Pencil work, no time pressure
Mixed below/above same base	Factors near <i>different</i> bases

The honest truth: Most multiplication problems in life are NOT Nikhilam-shaped. Long multiplication remains your everyday workhorse.

But when both factors *are* near a base — 97×96 , 1003×1004 — Nikhilam is 5–10× faster.

Tirthaji's other multiplication sūtra, *Ūrdhva-tiryagbhyām* ("vertically and crosswise"), works for any pair of numbers and is taught in its own module. Nikhilam is the *near-base specialist*; *Ūrdhva* is the *generalist*.

Homework

- Find 5 multiplication problems in your school textbook (any subject). For each, decide: Nikhilam, long, or *Ūrdhva*. Write 1 sentence justifying your choice. Bring tomorrow.

Differentiation

- Tier up:** Try a sub-base (base 50). Compute 47×48 via base 50: deficits 3, 2. Cross: $47 - 2 = 45$. Multiply by 50: 2250. Add deficit product $6 \rightarrow 2256$. (Real: $47 \times 48 = 2256$. ✓) Compare with long multiplication. Which felt faster?
- Tier down:** Stick to clear-cut cases. Distinguish: "is this near 100?" If yes, Nikhilam. If no, long.

Teacher notes

- This is the honesty day. Some students will be disappointed that Nikhilam isn't universally faster. That's the point: claims about Vedic mathematics often overstate the universal advantage. The honest framing protects against later disillusionment.
- The "decide" step before solving is the habit-of-mind worth building. Without it, students apply Nikhilam to inappropriate problems and get slower than the standard algorithm.
- The mention of *Ūrdhva-tiryagbhyām* sets up future modules. Don't elaborate on the procedure today — it's a 2-week topic of its own.

Citation(s) used in this lesson

- Vedic Mathematics Batch1.pdf* (Vedic Cultural Center, Sammamish WA) — Nikhilam scope and limits.
- Microsoft Word - sutras1.pdf* — sūtras #2 (Nikhilam) and #3 (*Ūrdhva-tiryagbhyām*).

Day 9 — Speed Sprints + Project Start

Module: Nikhilam Multiplication · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will demonstrate sub-10-second mental Nikhilam multiplication on Day 1's baseline problem (and similar) AND will have a project plan/draft ready for Day 10's showcase.

Materials

- Day 1 baseline sheet (per student) — they pull it from their workbook.
- Sprint sheet: 10 Nikhilam-suitable problems, timer-friendly.
- Project planning slip.
- `activities/activity-02-speed-sprint.md` worksheet.

Lesson flow

Warm-up (5 min)

- Complement-to-100 + excess-from-100 mixed round.
- “*Today is your retest day. Get sharp.*”

Speed sprint (15 min) — [activities/activity-02-speed-sprint.md](#)

Brief: - 10 Nikhilam problems on a sheet, mixed (both-below, both-above, mixed-sign, near-1000). - 3 minutes individual. Solve as many as you can. - 1 minute self-correction. - Repeat (round 2): swap problems on a fresh sheet, 3 minutes again. - 1 minute self-correction. - 3-minute paired chorus round: partner reads, the other answers in chorus.

Day 1 retest (5 min)

- Students pull their Day 1 baseline sheet ($97 \times 96 = 9312$).
- On a fresh strip, redo the problem using Nikhilam.
- Record new time.
- Compare: **Day 1 (long) vs. Day 9 (Nikhilam)**.

Most students will go from 30–60 sec → under 10 sec. Some will hit 3 sec.

Project planning (15 min)

- Distribute project planning slip (see [assessments/project-brief.md](#)).
- Each pair fills out:
 - Format chosen (video / workbook / data project).
 - One worked example they’ll feature.
 - Materials needed.
 - Who does what.
- Teacher circulates, gives quick approval on viability.

Wrap-up (5 min)

- Brief share-out: 3 pairs announce their plan in one sentence.
- Reminder: project must demonstrate the *Nikhilam* method correctly; must cite at least one source PDF.

Student-facing content

Day 1 retest:

My time on 97×96 (Day 1, long multiplication): ____ seconds My time on 97×96 (Day 9, Nikhilam): ____ seconds **My speedup:** ____×

Sprint scoring: 10 problems in 3 minutes. Each correct = 1 point. Each wrong with right method = 0.5 (you knew the method). Each wrong with wrong method choice = 0.

Project plan (fill in): - Format: _____ - Featured example: _____
- Materials: _____ - Partner / role split: _____

Homework

- Finish project plan at home.
- Practise your featured example until you can present it without notes.
- 5 mental Nikhilam problems a day, every day, even after this module ends.

Differentiation

- **Tier up:** Sprint round of 20 problems in 3 minutes (full speed).
- **Tier down:** Sprint sheet with 5 problems in 3 minutes (slower pace, still timed).

Teacher notes

- The retest moment is the most powerful one of the module. Make space for it — even quiet students will smile when they see their improvement.
- Some pairs won't have a clear project plan by end of class. Pull them aside for 5 min after — better to spend the time now than have a shaky project on Day 10.
- Confirm the data-project pairs have at least 5 willing test subjects lined up.

Citation(s) used in this lesson

- *Vedic Mathematics Batch1.pdf* — sprint problem source.
- See `assessments/project-brief.md` for the full project specification.

Day 10 — Final Assessment + Project Showcase

Module: Nikhilam Multiplication · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will demonstrate mastery of the Nikhilam multiplication method via a summative quiz and a 3-minute project presentation.

Materials

- Summative quiz (`quizzes/summative.md`).
- Stopwatch.
- Rubric per pair (`assessments/rubric.md`).
- Reflection slip per student.

Lesson flow

Warm-up (2 min)

- Brief: “Quiz first (15 min), presentations after.”

Part 1: Summative quiz (15 min)

- Distribute quizzes/summative.md. Closed notebook.
- Quiz covers:
 - Complement / excess fluency (timed).
 - Both-below-100 (2 problems).
 - Both-above-100 (1 problem).
 - Mixed-sign (1 problem).
 - Near-1000 (1 problem).
 - Decide-which-method (3 problems).
 - One short-answer “why does it work” (algebra).
- Collect at 15-min mark.

Part 2: Project presentations (25 min)

- 3 min presentation + 1 min Q&A per pair (4 min total).
- ~6 pairs fit in 25 min. **Reserve next class for remaining pairs** if needed.
- During each:
 - Teacher marks the rubric in real-time.
 - One assigned audience member asks one question.
 - Other audience members fill “I learned / I wonder” slips.

Wrap-up (3 min)

- Closing question (oral): “*What’s one thing you’ll keep using from Nikhilam? And one thing you’ll keep using long multiplication for?*”
- Reflection slip submitted.

Student-facing content

Today: quiz + presentations

15 min for the summative quiz (closed book).

3-minute presentation per pair, then 1-min audience question.

Reflection (due at end of class): 1. What did you set out to make/learn, and what did you actually end up with? 2. What’s the most surprising thing you discovered while making this? 3. Where will you use Nikhilam in your life outside this classroom?

Homework

- None. Module is complete.

Differentiation

- **Tier up:** Students finishing the quiz early can write a 1-paragraph “what would Module 11 look like for Grade 4?” — input for Primary band.
- **Tier down:** Open-notebook quiz for students with documented learning needs.

Teacher notes

- Mark quiz immediately after class while presentations are fresh.
- If presentations spill into Day 11, protect that time.
- The “reflection on Nikhilam in life” question (oral) — listen for genuine answers (“I’ll use it on price tags in the shop”; “I’ll use it for mental estimation”). The honest answer “I won’t use it much” is also valid.

Citation(s) used in this lesson

- All sources verified across days 1–9.

Writing Prompts

Writing Prompts — Module 11 Nikhilam Multiplication (Middle)

Six prompts. Useful as exit tickets, homework reflections, or project warm-ups.

Reflection prompts

- 1. (Day 1, end of class, 3 min)** When you saw the teacher solve 97×96 in three seconds, what did you feel? Curiosity, suspicion, “that can’t be right”? Be honest — describe the feeling.
- 2. (Day 9, end of class, 5 min)** Compare your Day 1 time on 97×96 (long multiplication) with your Day 9 time (Nikhilam). Has your *feeling* about mental math changed in the last two weeks? Be honest — even “no” is a valid answer.

Explanatory prompts

- 3. (Day 3 or Day 6, homework, ~120 words)** Explain the Nikhilam method to a Grade 4 student who has never seen it. No more than 6 sentences. Use one specific example. Don’t say “magic” — say what’s actually happening.
- 4. (Day 7, homework, ~150 words)** Explain the algebraic identity $(B - a)(B - b) = B(B - a - b) + ab$ in your own words. Use one numerical example. Show how each piece of the identity matches a step in the Nikhilam procedure.

Argumentative prompt

- 5. (Day 8 or homework, 1 paragraph)** Some say “Vedic mathematics” should be a mandatory subject in every Indian school. Others say it should be taught alongside the standard algorithm with honest historical framing. Pick a position and defend it in 5–7 sentences. Use what you’ve learned about (a) when Nikhilam is faster, (b) when long multiplication is faster, and (c) the honest history of Tirthaji’s 1965 book.

Creative prompt

6. (Day 5 or Day 8, optional, 1 page) Write a short script (50–150 words) in which a frustrated student and a confident Nikhilam user have a conversation. The frustrated student raises one good objection to the Nikhilam method (“but most multiplication problems aren’t near 100!”). The confident user gives a clear, honest answer. (No “and then they magically agreed” endings — let the disagreement be real.)

Prompt-to-band notes

- For **Primary** band, prompt 1 works as a drawing or oral response.
- For **Senior** band, prompt 5 deserves a full essay treatment with citations from `sources.md`.

How to give feedback

- Prompts 1, 2: acknowledge with one sentence — no correction.
- Prompts 3, 4: respond with “the part that landed” and “the part that confused me.”
- Prompt 5: use the rubric — does the writing acknowledge the other side, give a specific example, reach a clear position?
- Prompt 6: no grading. Read the best ones aloud (with permission).

Homework

Homework — Module 11 Nikhilam Multiplication (Middle)

Day 1 → Day 2

Task: Practice the deficit-from-10 reflex. Write the deficit of each from 10: $7 \rightarrow \underline{\quad}$; $8 \rightarrow \underline{\quad}$; $9 \rightarrow \underline{\quad}$; $6 \rightarrow \underline{\quad}$; $5 \rightarrow \underline{\quad}$; $4 \rightarrow \underline{\quad}$; $3 \rightarrow \underline{\quad}$; $2 \rightarrow \underline{\quad}$; $1 \rightarrow \underline{\quad}$.

Do all 9 in under 10 seconds. Repeat 3 times.

Time: 5 min.

Day 2 → Day 3

Task: Solve 10 single-digit $\times$ single-digit problems (both 6–9) using Nikhilam. Self-check against your times table. Specifically: 7×8 , 8×9 , 9×9 , 6×7 , 6×9 , 7×7 , 8×8 , 6×8 , 9×7 , 9×8 .

Time: 10 min.

Day 3 → Day 4

Task: Solve 10 problems where both factors are between 85 and 99. Write deficits, left, padded right, and answer for each.

Time: 15 min.

Day 4 → Day 5

Task: Mixed sheet — 10 problems, half both-below-100 and half both-above-100. Decide the case before solving.

Time: 15 min.

Day 5 → Day 6

Task: Bring a list of any 5 multiplication problems from your daily life (price tags in a shop, distances, anything). For each, decide: Nikhilam-suitable or not? Justify in 1 sentence.

Time: 10 min.

Day 6 → Day 7

Task: Submit your project plan paragraph in writing (4–6 sentences). Choose format. Choose featured example.

Time: 15 min.

Day 7 → Day 8

Task: Write a 1-paragraph explanation suitable for a Grade 7 student of the algebraic identity $(B-a)(B-b) = B(B-a-b) + ab$. Include one worked example with the algebra shown.

Time: 15 min.

Day 8 → Day 9

Task: Practise your featured project example until you can present it without notes. Bring ALL project materials tomorrow.

Time: 10 min.

Day 9 → Day 10

Task: Final touches on your project. Make sure you have: the artefact, your script/notes, your 3-min presentation timed in your head, your citation visible on the artefact.

Time: 15 min.